Problem Set 2

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Due: February 18, 2026

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in `R`, please include the code you used to get your answers. Please also include the `.R` file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub in `.pdf` form.
- This problem set is due before 23:59 on Wednesday February 18, 2026. No late assignments will be accepted.

We're interested in what types of international environmental agreements or policies people support (Bechtel and Scheve 2013). So, we asked 8,500 individuals whether they support a given policy, and for each participant, we vary the (1) number of countries that participate in the international agreement and (2) sanctions for not following the agreement.

Load in the data labeled `climateSupport.RData` on GitHub, which contains an observational study of 8,500 observations.

- Response variable:
  - `choice`: 1 if the individual agreed with the policy; 0 if the individual did not support the policy
- Explanatory variables:
  - `countries`: Number of participating countries [20 of 192; 80 of 192; 160 of 192]
  - `sanctions`: Sanctions for missing emission reduction targets [None, 5%, 15%, and 20% of the monthly household costs given 2% GDP growth]

Please answer the following questions:

1. Remember, we are interested in predicting the likelihood of an individual supporting a policy based on the number of countries participating and the possible sanctions for non-compliance.

   Fit an additive model. Provide the summary output, the global null hypothesis, and *p*-value. Please describe the results and provide a conclusion.

   My Answer:

   Fit an additive model and provide summary output (Table 1).

   I changed the variables from ordered to unordered factors. For the countries variable, this made "20 of 192" the baseline for number of participating countries, while "None" was the baseline for the sanctions for missing emission reduction targets variable.

```
# Changing the variables to not be ordered factors and instead unordered factors
climateSupport$countries <- factor(climateSupport$countries, ordered = FALSE)
climateSupport$sanctions <- factor(climateSupport$sanctions, ordered = FALSE)

# Run additive model
add_model <- glm(data = climateSupport,
                 choice ~ countries + sanctions,
                 family = binomial)

summary(add_model)
```

Table 1: Table of Coefficients

	Dependent variable:
	Choice Additive Model
Countries (80 of 192)	0.336^{***}
	(0.054)
Countries (160 of 192)	0.648^{***}
	(0.054)
Sanctions (5%)	0.192^{***}
	(0.062)
Sanctions (15%)	-0.133^{**}
	(0.062)
Sanctions (20%)	-0.304^{***}
	(0.062)
Constant	-0.273^{***}
	(0.054)
Observations	8,500
Log Likelihood	−5,784.130
Akaike Inf. Crit.	11,580.260
Note:	*p<0.1; **p<0.05; ***p<0.01

Global null hypothesis, p-value, results discussion, and conclusion.

The global null hypothesis states that neither of the variables, number of participating countries and sanctions for missing emission reduction targets, are significant in predicting whether or not an individual agrees with the given policy. This means that at no levels of either variable would they be significant in predicting the likelihood a person supports a policy. The alternative hypothesis states that at least one of the variables listed, at any level, is significant in predicting the likelihood of whether or not an individual agrees with the given policy.

H_0: $\beta_{countries} = \beta_{sanctions} = 0$, H_a : At least one $\beta \neq 0$

The additive model that was run above can be tested against the null model to determine whether or not the variables, and specifically their different levels, are significant in predicting the likelihood of an individual supporting a policy. A likelihood ratio test will be utlilized to test the log-likelihoods of the two models against one another.

```r
# Run LRT on the null vs additive model
# Run Null model
null_model <- glm(data = climateSupport,
                  choice ~ 1,
                  family = binomial)

# Run test on null vs additive model
anova(null_model, add_model, test = "LRT")
```

```
Analysis of Deviance Table

Model 1: choice ~ 1
Model 2: choice ~ countries + sanctions
Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
1      8499      11783
2      8494      11568  5   215.15 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

In Table 1, it can be seen that the coefficients for the numbers of participating countries, as well the possible sanctions are significant. This means that they each have a significant impact on predicting the likelihood of an individual supporting a policy. This finding can be further reinforced by the results from the likelihood ratio test between the null model, where no coefficients are present, and the additive model, where all coefficients are present. The p-value for this test was less than 2.2e-16, indicating that the null hypothesis that the number of participating countries and amount of sanctions for missing emission reduction targets are not significant can be rejected, meaning at least one of these variables is signifiant in predicting the likelihood of support. Again, this is also seen in the regression output table for the additive model, as all of the coefficients are noted as significant in their predictive power and should subsequently be included in the model.

2. If any of the explanatory variables are statistically significant in this model, then:

   (a) For the policy in which nearly all countries participate [160 of 192], how does increasing sanctions from 5% to 15% change the odds that an individual will support the policy? (Interpretation of a coefficient)

   As an additive model has been fitted, the number of participating countries does not have an impact on the change in odds when increasing sanctions from 5% to 15%. Looking solely at the sanctions variable, it is in reference to the "None" level, which is the baseline, so the difference between the coefficients for 5% and 15% sanctions must be mathematically compared against one another.

```r
# 2a: Determining the log-odds difference between 5% and 15% sanctions.
diff_5_15 <- coef(add_model2)["sanctions15%"] - coef(add_model2)["sanctions5%"]
```

```
-0.3251028
```

   This indicates that in accordance with our additive model, when holding all other variables constant, when the sanctions are increased from 5% to 15%, the log odds that an individual will support the policy decrease by 0.325. The odds ratio can also be computed for this, using the following formula:

$$OddsRatio = e^{\beta_j}$$

   In the case of our β_j, the log-odds difference between $\beta_{sanctions5}$ and $\beta_{sanctions15}$ of -0.325 will be utilized.

```r
# Determining odds ratio between 5% and 15% sanctions
odds_ratio_5_15 <- exp(diff_5_15)
```

```
0.7224531
```

   Using the odds ratio, we can interpret the odds that an individual will support a policy when sanctions are increased from 5% to 15% as follows: the log-odds that an individual will support a policy decrease by 72.25% when the sanctions are increased from 5% to 15%.

(b) For the policy in which very few countries participate [20 of 192], how does increasing sanctions from 5% to 15% change the odds that an individual will support the policy? (Interpretation of a coefficient)

As this is an additive model, the number of participating countries does not have an impact on the change in odds when increasing sanctions from 5% to 15%. To interpret the change in odds for increasing the sanctions, it will be the same at all levels of country participation and is therefore the same as in question 2(a). When sanctions are increased from 5% to 15%, and all other variables are held constant, the log-odds that an individual will support the policy decrease by 0.325, or 72.25%.

(c) What is the estimated probability that an individual will support a policy if there are 80 of 192 countries participating with no sanctions?

The estimated probability that an individual will support a policy if there are 80 out of 192 countries participating with no sanctions can be determined by solving for the log-odds and placing this into a formula to determine the probability that y = 1. The formula for this is as follows:

$$\pi_i = P(Y_i = 1|X_i) = \frac{exp(\beta_0 + \beta_1 X_{1i} + ...\beta_k X_{ki})}{1 + exp(\beta_0 + \beta_1 X_{1i} + ...\beta_k X_{ki})}$$

The coefficients for β_0 and β_1 can be input and utilized to solve for π_i, which is the probability that $Y_i = 1$ when β_1 represents the effect on the log-odds that an individual will support a policy if there are 80 of 192 countries participating.

```r
# 2c: Estimated probability an individual will support the policy if
    there are
# 80 out of 192 countries participating.
probability <- exp(coef(add_model2)["(Intercept)"] +
                   coef(add_model2)["countries80 of 192"])/
  (1 + exp(coef(add_model2)["(Intercept)"] +
               coef(add_model2)["countries80 of 192"]))
```

```
0.5159191
```

The estimated probability that an individual will support a policy if there are 80 of 192 countries participating with no sanctions is 0.5159, or 51.59%.

3. Would the answers to 2a and 2b potentially change if we included an interaction term in this model? Why?

Yes, the answers to 2a and 2b would potentially change if we included an interaction term in this model. This is due to the fact that we would no longer have an additive model and would now have an interaction model. An interaction model would mean that there is an interaction effect between the explanatory variables, such that a change in one variable impacts how the other variable affects the log-odds of the outcome variable. In this specific data, an interaction term would mean that the number of countries participating impacts the effect of how the sanctions for missing reduction targets affects the log-odds an individual supports a policy, or vice versa. For this reason, the answers to 2a and 2b would potentially change, as the change in odds that an individual supports the policy due to increasing sanctions from 5% to 15% could be different at differing levels of country participation due to the interaction between country participation and sanctions.

- Perform a test to see if including an interaction is appropriate.

```r
# Run interaction model
int_model <- glm(data = climateSupport,
                 choice ~ countries * sanctions,
                 family = binomial)
```

```r
# Perform test on additive vs interaction model
anova(add_model, int_model, test = "LRT")
```

```
Analysis of Deviance Table

Model 1: choice ~ countries + sanctions
Model 2: choice ~ countries * sanctions
Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1      8494      11568
2      8488      11562  6   6.2928   0.3912
```

From the analysis of deviance table, and specifically the p-value of 0.3912, it can be determined that including an interaction is not appropriate for this data. The p-value is larger than 0.05, indicating that the additive model is sufficient in its explanation of the likelihood that someone supports the policy.